

PAVAN KUMAR P H V

Senior Applied AI Engineer · Forward-Deployed AI Engineering

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PROFILE

AI engineer and consultant with five years of production experience, spanning enterprise analytics, agentic AI, backend services, real-time computer vision and embedded deployment. Work covers AI orchestration, data integration, KPI reconciliation, retrieval and observability. M.Tech in Data Science, with cloud and AI architecture certifications.

CORE EXPERTISE

AI & retrieval: LangGraph, LangChain, RAG, semantic search, hybrid vector and SQL retrieval, embeddings, evaluation and Langfuse tracing.

Backend & data: Python, FastAPI, Flask/Gunicorn, REST APIs, asynchronous jobs, PostgreSQL, pgvector, Redis, MongoDB, SAP Datasphere/OData.

Deployment & cloud: Docker, Kubernetes, CI/CD, Azure, GCP and AWS.

Computer vision: YOLOv7/v9, TensorRT, ONNX, LightGlue, RGB-D reconstruction, binocular vision and embedded model optimisation.

PROFESSIONAL EXPERIENCE

Kearney

2026 – present

Senior AI Consultant — Analytics & Information Management

Bangalore, India

Enterprise analytics & AI engineering

- Contributed AI orchestration, backend services and enterprise data integration to team-built analytics systems.
- Worked on KPI lineage and reconciliation, testing and observability, using Python, FastAPI, LangGraph, SAP data services and Langfuse.

Process knowledge & retrieval

- Engineered backend and knowledge-repository capabilities for process documentation and searchable organisational knowledge.
- Contributed asynchronous orchestration, semantic chunking and embedding, and hybrid vector and governed SQL retrieval.

Solum

Aug 2024 – 2026

AI/CV & ML Systems Engineer

Bangalore, India

- Designed multi-camera retail shelf-vision pipelines using YOLOv7/v9, TensorRT and LightGlue, delivering sub-200 ms responses at 15–20 FPS.
- Built hot-reload RFID ML-serving APIs with Flask/Gunicorn, Azure Event Hubs and MongoDB, with containerised deployment and observability.
- Built LLM-powered log investigation using fuzzy clustering, ReAct-based LangChain agents and locally hosted LLaMA models.

Ignitarium

Aug 2022 – Jul 2024

AI Engineer

Bangalore, India

Edge AI & industrial vision

- Led architectural refinement of FastestDet, improving accuracy by 35%, and deployed real-time detection models on Renesas microcontrollers.
- Designed RGB-D reconstruction and binocular-vision pipelines for industrial defect detection.
- Developed domain-specific LLM Q&A systems for railway, telecom and solar applications.

Selected embedded-vision engagement · Feb 2023 – Dec 2023

- Built and optimised custom object-detection and segmentation models for embedded devices, including real-time deployment on IMX500 intelligent vision sensors.
- Used transfer learning to reduce training time and integration effort.

Ignitarium

Jun 2021 – Jul 2022

AI Intern

Bangalore, India

- Developed and evaluated machine-learning and deep-learning prototypes across computer vision, model optimisation and deployment-oriented experimentation.

EARLIER EXPERIENCE

Freelance teaching assistant

Aug 2020 – Dec 2021

Teaching assistance alongside postgraduate study in Coimbatore.

EDUCATION

M.Tech, Data Science

2022

Amrita School of Engineering, Coimbatore

B.Tech, Mechanical Engineering

2019

Amrita School of Engineering, Coimbatore

PROFESSIONAL CERTIFICATIONS

- **Professional Cloud Architect — Google Cloud (2026)**
[Verify credential on Credly](#)
- **Claude Certified Architect – Professional — Anthropic (2026)**
[Verify credential on Credly](#)

PORTFOLIO

phvpavankumar.github.io — selected engineering work and illustrative demonstrations.

Client work is confidential. Portfolio demonstrations use fictional data and simplified workflows; they are not employer or client products.